

The four leakage smells

If you see one of these, the accuracy number on the slide is not real. Point at this poster.

why it is a leak

the fix

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```
train_test_split(  
    shuffle=True)
```

Test rows end up preceding training rows. The model is scored on hours whose future it has seen.

Split by TIME. Always.
There is no exception for 'just a quick check'.

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```
rolling(w,  
    center=True)
```

The window spans $[t-12, t+11]$ for $w = 24$ — it swallows the target hour and its neighbours.

`y.shift(1).rolling(w, center=False)`
The window must END before t .

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```
scaler.fit(  
    all_data)
```

The mean and standard deviation of the future are subtracted from the past.

Fit every scaler, encoder and imputer on the training slice only, inside the backtest loop.

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```
exog =  
    realised_temp
```

At 14:00 you do not have tomorrow's realised temperature. You have a forecast of it.

Use the FORECAST of the covariate, and let its error widen your interval honestly.

Every feature, every split and every scaler must be computable using only data that existed at the forecast origin.

ADF and KPSS have opposite null hypotheses

ADF H_0 = unit root $\rightarrow$ a small p means STATIONARY

KPSS H_0 = stationary $\rightarrow$ a small p means NON-STATIONARY

KPSS says stationary
p > 0.05

KPSS says non-stationary
p < 0.05

**ADF says
stationary**
p < 0.05

STATIONARY
proceed — no differencing

DIFFERENCE-STATIONARY
difference once, then re-test

**ADF says
non-stationary**
p > 0.05

TREND-STATIONARY
detrend, or add a trend
regressor — do not difference

NON-STATIONARY
difference and re-test

Difference as little as possible. Seasonal first, then regular. Over-differencing injects artificial negative lag-1 autocorrelation.

Which metric, and what it hides

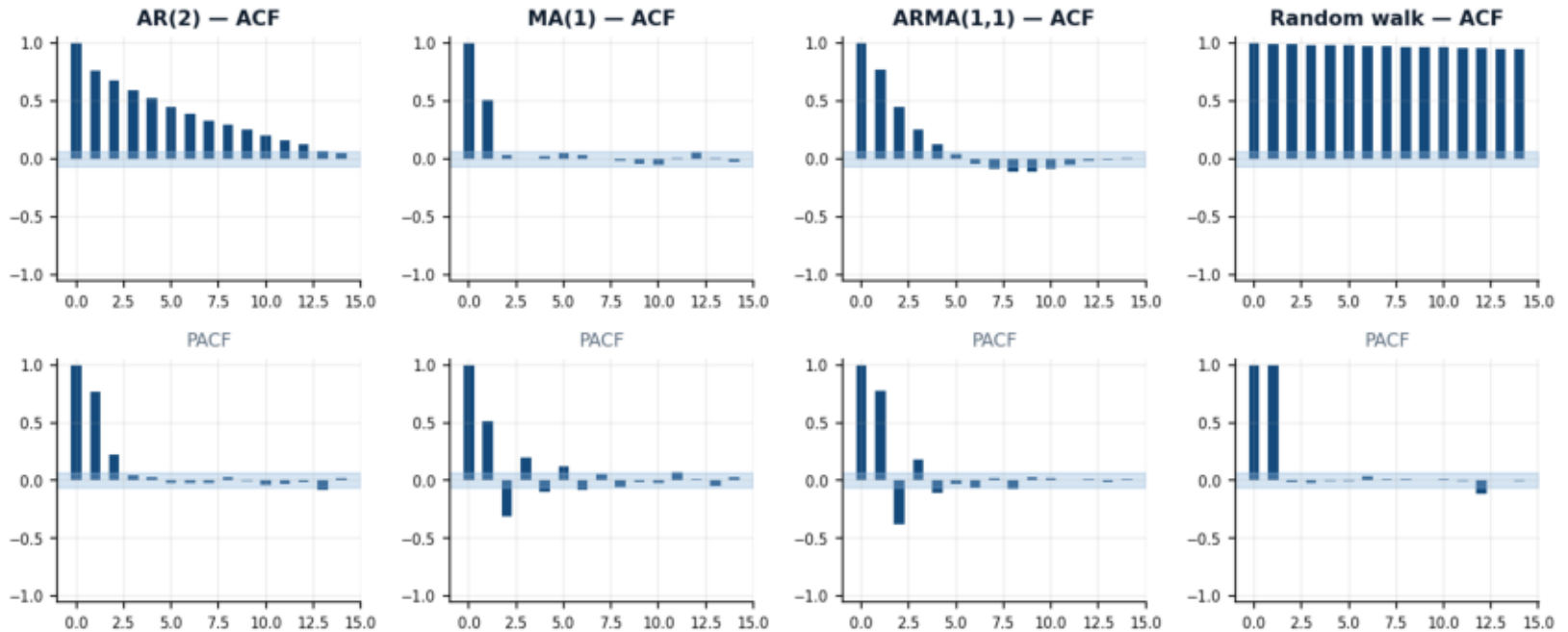
Report MASE alongside a percentage metric. The percentage is what a planner feels; the MASE is what makes it comparable.

metric	formula	strength	fails when
MAE	mean $ y - \hat{y} $	same units, robust	not comparable across series of different scale
RMSE	$\sqrt{\text{mean } (y - \hat{y})^2}$	penalises large misses	dominated by outliers; scale-bound
MAPE	mean $ y - \hat{y} / y $	scale-free, intuitive	explodes near zero; over-forecasts capped at 100%
sMAPE	symmetric percentage	bounded	still unstable near zero; awkward to interpret
MASE	MAE / MAE of seasonal-naive	scale-free, symmetric, defined at zero	at zero a sensible baseline period m
Pinball	the quantile loss, averaged	a PROPER score — rewards honest calibration	not useful for a probabilistic forecast
Coverage	$P(y \text{ inside the interval})$	the number operations feels	marginal coverage hides conditional holes

MASE < 1 beats the free seasonal-naive baseline · MASE = 1 ties it · MASE > 1 means you should have shipped the baseline.

The correlogram fingerprint

AR → PACF cuts off · MA → ACF cuts off · ARMA → both tail off · slow linear decay → difference before reading further



PACF cuts off after lag 2
→ AR(2)

ACF cuts off after lag 1
→ MA(1)

both tail off
→ ARMA; use AICc/BIC

ACF decays slowly and linearly
→ NOT stationary. Difference first.

"Cuts off" means it drops inside the $\pm 1.96/\sqrt{n}$ band. The default lags=20 never reaches lag 24 — set lags $\geq 2m$ or the seasonal spike is invisible.

Every series is three things at once

The shared vocabulary between the data scientist and the decision-maker.

$$y_t = T_t + S_t + R_t \quad y_t = T_t \times S_t \times R_t$$

additive

multiplicative $\Leftrightarrow \log y = \log T + \log S + \log R$

T_t trend

the long-run level — growth, decline, a structural break

S_t seasonality

the repeating pattern of KNOWN period: 24, 168, 8766

R_t remainder

what neither explains — and where a missing covariate hides

**Decide on EVIDENCE: correlate the per-cycle level with the per-cycle amplitude.
If the swing grows with the level, the series is multiplicative — model $\log(y)$.**